Correlated Topic Model for Image Annotation

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***Abstract*—For the task of image annotation, traditional methods based on probabilistic topic model, such as correspondence Latent Dirichlet Allocation (corrLDA) [1], assumes that image is a mixture of latent topics. However, this kind of models is unable to directly model correlation between topics since topic proportions of an image are generated independently. Our model, called correspondence Correlated Topic Model (corrCTM), extends Correlated Topic Model (CTM) [2] from natural language processing to capture topic correlation from covariance structure of more flexible model distribution. Unlike previous LDA based models, topic proportions are correlated with each other in proposed corrCTM. And the topic correlation propagates from image features to annotation words through a generative process, and finally correspondence between images and words could be generated. We derive an approximate inference and estimation algorithm based on variational method. We examine the performance of our model on two benchmark image datasets, show improved performance over corrLDA for both annotation and modeling word correlation.**

I. Introduction

Image annotation is an important open problem in computer vision research. Given an image with no caption, the goal of image annotation is to assign relevant text keywords from a word vocabulary to the image, to describe its visual content. Image annotation can be applied in object recognition (propose keywords for image), or text-based image retrieval (propose images from a combination of keywords). A lot of attention in recent years has been paid to the design and development of automatically annotating images.

Generally, an annotation algorithm is to learn patterns of association between image and text from a set of annotated images, then predict textual annotations for any unlabeled image. Previous approaches can be mainly categorized into 2 types.

For the first type, a classifier is learned for each vocabulary word as which is treated as "*concept class*." [3] This is a supervised learning model, and often SVMs or boosting classifiers are typically used. In practice, this line of approaches would inevitably suffers from scalability problem, as the annotation performance sharply reduces when the discriminative boundaries for large scale of classifiers become ambiguous.

In the case of second type, the joint probabilistic distribution between image feature and text word is learned by a generative latent factor framework [1] [4] [5], which assumes that each image is formed with a small set of hidden factors, and these factors correspondingly associate with text words. Compared with the first type, this generative approach learns a single model for all vocabulary words and yields a better modeling of image and word dependencies.

With regard to the specific case of probabilistic topic model, which is a sub-branch of the second type, the concept of *latent topic* is introduced to reduce original higher image feature space and text word space to lower topic space, then unsupervised method or supervised method is used to build the relationship between image feature and text word.

For unsupervised method, the pioneering work of correspondence LDA (corrLDA) proposed by Blei *et al.*, takes topic representation for image feature and text word, and correspondingly associates text word with one of the image feature. Later, Wang *et al.* [4] designs a simultaneous classification and annotation framework which extends from corrLDA, allows image feature and text word share same dimensional topic space. Putthividhya *et al.* [6] proposes a more general and flexible annotation model which allows different topic spaces for image feature and text word.

For supervised method, text words are treated as multivariate response variables given the topic space of image feature. Then correspondence between topic space and response variable can be obtained by regression model, different distribution can be assumed as response variable, such as Bernoulli [7], softmax [8].

Although models above have made improvement for the image annotation task, some intrinsic issues should not be neglected. Firstly, all these models are extend from LDA, which is based on the assumption that topic proportions for each image are generated from Dirichlet distribution. Under a Dirichlet, proportions of each topic are nearly independent, which indicates that topics share nearly no correlations. Secondly, topic independency is transferred to visual features and annotation words in original LDA model, which implies visual features and annotation words themselves have no correlations.

Indeed, images with similar visual content would share similar visual features, topic proportions, and co-occurred annotation words. Inspired by Correlated Topic Model (CTM) [2], which uses more flexible covariance structure to capture topics' correlation in text corpora, we propose a corresponding correlated topic model (corrCTM). Two main contributions of our work are:

- We make the first trail to extend CTM to image annotation problem, and develop a generative model

to capture the relationship between images and words through topic correlation.

- We derive a mean-field variational inference algorithm for approximate posterior inference in the corrCTM model.

Through experimental evaluation on benchmark image annotation datasets, we find that the proposed corrCTM not only improves the annotation performance, but also better structures the visual feature co-occurrence and annotation word co-occurrence.

This paper is organized as follows. In Section 2, we describe our proposed model and generative process using graphical representation. In section 3, we derive a variational algorithm for parameter estimation, inference and prediction. In Section 4, we evaluate the ability of proposed method in capturing word co-occurrence, and present experimental results on image annotation tasks. Finally, we summarize our finding in Section 5 and make conclusion.

II. Proposed Model

In this section, we first give data representation and notations generally used by topic models [1] [4] [6]. Then, we depict the generative process of proposed model by graphical representation.

A. Data Representation and Notation

We follow the instruction from statistical text document analysis, and each image is represented as a bag of "codewords". Given a training set of images with annotation words, we use the following notation. Each image is a collection of M visual feature codewords, denoted as $\mathbf{v} = \{v_{1:M}\}$, where v_m is a unit-basis vector of size V_s with exactly one non-zero entry representing the index of current visual feature in the visual feature dictionary of size V_s. Similarly, for one image annotated with N words $\mathbf{w} = \{w_{1:N}\}$, we denote each word w_n as a unit-basis vector of size V_t again with only one takes value 1 and 0 otherwise, here V_t is the word dictionary size. Thus, a collection of D training image-word pairs can be denoted as $\{\mathbf{v}_{1:D}, \mathbf{w}_{1:D}\}$.

B. Correspondence Correlated Topic Model (corrCTM)

The proposed corrCTM is built on CTM. Compared with LDA model, the CTM makes two modifications: 1) using multivariate Normal distribution instead of Dirichlet to capture the topic correlation, 2) adding an additional logistic normal mapping step to incorporate the normally distributed topics. These modifications make CTM give a better fit than LDA in representing textual collections. Furthermore, Malisiewicz *et al.* [9] has applied CTM to object recognition in image area and declared that CTM also has a better fit for visual representation than LDA by modeling correlated topic space rather than independent topic space.

Motivated by such a success of CTM in recognition task, here we adopt the association model of CTM for the task of annotation. In order to capture the correlation of topics, we model the hyper parameter of topic prior distribution as multivariate Normal distribution instead of Dirichlet, and structure topics' dependencies by covariance matrix, and then we use the logistic normal function $f(\theta_i) = \left(\exp\ \theta_i / \sum_{j=1}^{K} \exp\ \theta_j\right)$ to project the multivariate normal to topic proportions for each image, here K is the topic number.

Let $\{\mu, \Sigma\}$ be a K-dimensional mean and variance matrix from normal distribution, and let topics $\pi_{1:K}$ be K multinomials over a fixed visual feature vocabulary with size V_s, $\beta_{1:K}$ be K multinomials over a fixed text word vocabulary with size V_t. The proposed corrCTM generates an image-word pair with M image codewords and N annotation words from the following generative process:

1) Draw topic proportions $\theta|\{\mu, \Sigma\} \sim \mathrm{N}(\mu, \Sigma)$
2) For each visual feature v_m, $m \in \{1, ..., M\}$:
   (a) Draw topic assignment $z_m \sim \mathrm{Mult}(f(\theta))$
   (b) Draw visual feature $v_m \sim \mathrm{Mult}(\pi_{z_m})$
3) For each textual word w_n, $n \in \{1, ...N\}$:
   (a) Draw feature index $y_n \sim \mathrm{Unif}(1, ..., M)$
   (b) Draw textual word $w_n \sim \mathrm{Mult}(\beta_{z_{y_n}})$

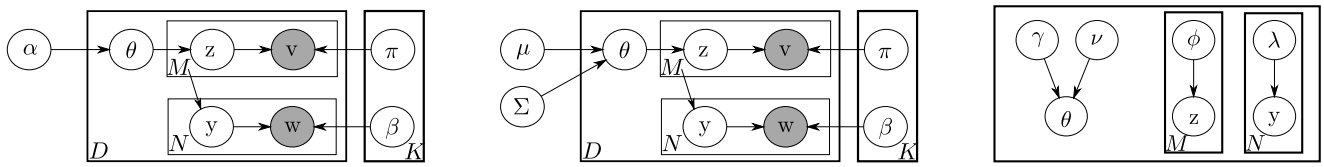
In particular, we first generate M image features v_m from correlated topic proportions θ, conditional on the topic-feature multinomial π, then for each of the N text words, one of the M features is selected and corresponding assigned to a text word w_n, conditional on the topic-word multinomial β. This model is shown as a directed graphical model in Fig. 1(b), and its generative process is identical to the process of corrLDA in Fig. 1(a), except that the topic proportions for the image are drawn from a logistic normal distribution rather than a Dirichlet.

From the generative process of corrCTM, it could be learned that topic correlations are modeled and generated through the covariance matrix Σ of prior multivariate normal distribution. After the logistic normal procedure, the topic proportions are generated depending on the correlations. And later topic assignment for image features and text words would again rely on the correlated mixture of topics in the current image.

III. Variational EM

To learn the paramters of corrCTM that maximizes the likelihood of training data, we employ the Expectation Maximization (EM) framework that iteratively estimates the model parameters of latent variables. Generally, in the E-step, we derive posterior probability function using Jensen's inequality to lower-bounds the data likelihood function, then perform to maximize the lower-boundary to calculate model parameters. Unfortunately, for most latent variable models, directly computing the exact posterior function over latent variables in the E-step is computationally intractable.

Thus, variational EM framework is widely used to approximate the inference instead of exact inference in the E-step, then maximize the lower-boundary to obtain a new model parameter estimate. In other words, the two alternative procedures in variational EM consists of variational inference and parameter estimation. The former E-step is to add and calculate a set of variational parameters to obtain the approximate lower-bound on likelihood of each sample. The latter M-step is to estimate the model parameters that maximize the log likelihood of the whole training samples.



(a) corrLDA (b) corrCTM (c) variational approximation for corrCTM

Fig. 1. Graphical representation of (a)original corrLDA, (b)proposed corrCTM, (c)variational distribution used to approximate the posterior for each image in corrCTM. Gray node represents observed variable, white node represents latent variable.

In this section, we first derive a variational inference algorithm for approximating the posterior distribution, and estimating model parameters. Next, we derive prediction algorithm for annotating new image with model parameters.

A. Variational Inference

In our corrCTM model, the latent variables are the per-image topic proportions θ, the per-codeword topic assignment z_m, and the per-annotation word index y_n. In the graphical representation of corrCTM in Fig. 1(b), θ is conditional dependent with model parameter π, which leads to intractability for computing the log likelihood. A simple way is to add a set of free variational parameters and modify the original graphical model, to ensure that latent variables are conditional independent, as illustrating in Fig. 1(c).

The mean-field variational distribution is

$$q(\theta, z, y) = q(\theta|\gamma, \nu) \prod_{m=1}^{M} q(z_m|\phi_m) \prod_{n=1}^{N} q(y_n|\lambda_n), \quad (1)$$

where (γ, ν) is variational mean and variance of normal distribution, ϕ_m is a variational multinomial over K topics, λ_n is a variational multinomial over codewords.

Let $\Delta = \{\mu, \Sigma, \pi, \beta\}$ denotes model parameters, following [10], we bound the log-likelihood of an image-annotation pair $(\mathbf{v}, \mathbf{w})$:

$$\begin{aligned} \log\ p(\mathbf{v}, \mathbf{w}|\Delta) &= \log \int_{\theta} \sum_{z,y} \frac{p(\mathbf{v}, \mathbf{w}, \theta, z, y|\Delta) q(\theta, z, y)}{q(\theta, z, y)} dz dy \\ &\geq \mathbb{E}_q[\log\ p(\mathbf{v}, \mathbf{w}, \theta, z, y|\Delta)] - \mathbb{E}_q[\log\ q(\theta, z, y)] \\ &= \mathcal{L}(\gamma, \nu, \phi, \lambda; \Delta), \end{aligned} \quad (2)$$

where $\mathbb{E}_q$ is the expectation according to the variational distribution. Take the log likelihood function $\mathcal{L}(\gamma, \nu, \phi, \lambda; \Delta)$ as object function, we fit these parameters with coordinate ascent to maximize the object function.

From Fig. 1(c), as all latent variable are now conditional independent, following the generative procedure, the object function $\mathcal{L}(\gamma, \nu, \phi, \lambda; \Delta)$ can be factorized as

$$\begin{aligned} \mathcal{L} \ = \ & \mathbb{E}_q[\log\ p(\theta|\mu, \Sigma)] \\ & + \sum_m (\mathbb{E}_q[\log\ p(z_m|\theta)] + \mathbb{E}_q[\log\ p(v_m|z_m, \pi)]) \\ & + \sum_n (\mathbb{E}_q[\log\ p(y_n|M)] + \mathbb{E}_q[\log\ p(w|y_n, z_m, \beta)]) \\ & - \mathbb{E}_q[\log\ q(\theta|\gamma, \nu)] \\ & - \sum_m \mathbb{E}_q[\log\ q(z_m|\phi)] - \sum_n \mathbb{E}_q[\log\ q(y_n|\lambda)]. \end{aligned} \quad (3)$$

The coordinate ascent updates for λ is the same as in [1], for ζ, γ, ν is the same as in [11], we summarize them using similar notation:

$$\hat{\lambda_{nm}} \propto \exp\ \left(\sum_{i=1}^{K} \phi_{mi} \log\ \beta_{i,w_n} \right), \quad (4)$$

$$\hat{\zeta} = \sum_{i=1}^{K} \exp\ \{\gamma_i + \nu_i^2/2\}, \quad (5)$$

$$\frac{\partial \mathcal{L}(\gamma)}{\partial \gamma} = -\Sigma^{-1}(\gamma - \mu) + \sum_{m=1}^{M} \phi_{m,1:K} - (M/\zeta) \exp\ (\gamma + \nu^2/2), \quad (6)$$

$$\frac{\partial \mathcal{L}(\nu_i)}{\partial \nu_i^2} = -\Sigma_{ii}^{-1}/2 - (M/2\zeta) \exp\ (\gamma + \nu_i^2/2) + 1/(2\nu_i^2). \quad (7)$$

Given the first derivation of γ, ν, Newton's method can be used to find the local optimal of the coordinate object function. In practice, we use LBFGS [12] and constrained Newton method to iterate γ and ν. ζ is an additional variational parameter which is introduced to lower-bound the expectation of logistic normal for topic assignment $\mathbb{E}_q[\log\ p(z_m|\theta)]$. Details could be referred in [11].

We next turn to update variational multinomial ϕ. Here, the variational method derived in [1] cannot be used because the expectation of the log partition function is logistic normal rather than Dirichlet. The terms in $\mathcal{L}$ with respect to ϕ_m are:

$$\begin{aligned} \mathcal{L}(\phi_m) \ = \ & \sum_{i=1}^{K} \phi_{mi} \Big(\gamma_i + \log\ \pi_{i,v_m} - \log\ \phi_{mi} \\ & + \sum_{n=1}^{N} \lambda_{nm} \log\ \beta_{i,w_n} \Big), \end{aligned} \quad (8)$$

as to the multinomial constrain $\sum_{i=1}^{K} \phi_{mi} = 1$, we can add Lagrange multiplier to $\mathcal{L}(\phi_{mi})$, and take derivative to ϕ_{mi}, we obtain:

$$\begin{aligned} \frac{\partial \mathcal{L}(\phi_{mi})}{\partial \phi_{mi}} \ = \ & \gamma_i + \log\ \pi_{i,v_m} + \sum_{n=1}^{N} \lambda_{nm} \log\ \beta_{i,w_n} \\ & - \log\ \phi_{mi} + C, \end{aligned} \quad (9)$$

where C is a constant. Setting $\partial \mathcal{L}(\phi_{mi}) / \partial \phi_{mi} = 0$ leads to

$$\hat{\phi_{mi}} \propto \pi_{i,v_m} \exp\ \left(\gamma_i + \sum_{n=1}^{N} \lambda_{nm} \log\ \beta_{i,w_n} \right). \quad (10)$$

Observe that how the per-codeword variational distribution over topics ϕ depends on the different topic proportions hyper

parameter γ_i of topic proportion. In corrLDA, γ_i, $i \in \{1, ..K\}$ are generated from Dirichlet, which implies that γ_i, γ_j, $i, j \in \{1, ..., K\}$ are nearly independent, however, in proposed corrCTM, the γ_i is generated from normal distribution, and γ_i, γ_j are calculated relied on each other in Equation 6, where the variance ν_i involved in the covariance matrix Σ. Thus, the covariance structure highly correlates topic proportions for each image, and the topic assignment for the codewords (ϕ_{mi}) and word (λ_{nm}) would also be correlated.

The full variational inference procedure repeats the updates of Equation 4, 6, 7, 5, 10, until Equation 3, the object function converges. Because the Equation 6 for γ and Equation 7 for ν are not close-form solution to their maximization, the variational inference in corrCTM is more computational time-consuming than corrLDA, which is the disadvantage of proposed corrCTM.

B. Parameter Estimation

Given a collection of image data with annotation words $\{(\mathbf{v}_d, \mathbf{w}_d)\}_{d=1}^{D}$, our objective is to find the maximum likelihood estimation for model parameters $\{\mu, \Sigma, \pi, \beta\}$. Recall $\Delta = \{\mu, \Sigma, \pi, \beta\}$, the overall log likelihood of collection is bounded by

$$\begin{aligned} \mathcal{L} &= \sum_{d=1}^{D} \log\, p(\mathbf{v}_d, \mathbf{w}_d|\Delta) \\ &\geqslant \sum_{d=1}^{D} \mathcal{L}_d(\gamma_d, \nu_d, \phi_d, \lambda_d; \Delta). \end{aligned} \tag{11}$$

Again, we maximize the lower boundary of $\mathcal{L}$ by plugging Equation 3 into Equation 11. We then update model parameters Δ by setting derivation equal zero with respect to each model parameter.

The terms containing $\pi_{1:K}$ (with Lagrange multipliers) are

$$\begin{aligned} \mathcal{L}(\pi_{ij}) &= \sum_{d=1}^{D}\sum_{m=1}^{M_d}\sum_{i=1}^{K}\sum_{j=1}^{V_s} \phi_{dmi} \log\, \pi_{i,v_m} \\ &\quad -C\left(\sum_{j=1}^{V_s} \pi_{ij} - 1\right). \end{aligned} \tag{12}$$

Setting $\partial\mathcal{L}(\pi_{ij})/\partial\pi_{ij} = 0$ leads to

$$\hat{\pi_{ij}} \propto \sum_{d=1}^{D}\sum_{m=1}^{M_d} \phi_{dmi} v_{dm}^{j}. \tag{13}$$

The terms containing $\beta_{1:K}$ (with Lagrange multipliers) are

$$\begin{aligned} \mathcal{L}(\beta_{i,j}) &= \sum_{d=1}^{D}\sum_{n=1}^{N_d}\sum_{i=1}^{K}\sum_{j=1}^{V_t}\sum_{m=1}^{M_d} \phi_{dmi}\lambda_{nm} \log\, \beta_{i,w_n} \\ &\quad -C\left(\sum_{j=1}^{V_t} \beta_{ij} - 1\right). \end{aligned} \tag{14}$$

Setting $\partial\mathcal{L}(\beta_{ij})/\partial\beta_{ij} = 0$ leads to

$$\hat{\beta_{ij}} \propto \sum_{d=1}^{D}\sum_{n=1}^{N_d} w_{dn}^{j} \sum_{m=1}^{M_d} \phi_{dmi}\lambda_{dnm}. \tag{15}$$

The term containing μ are

$$\mathcal{L}(\mu) = -(1/2)\sum_{d=1}^{D} (\gamma_d - \mu)^T \Sigma^{-1} (\gamma_d - \mu). \tag{16}$$

Setting $\partial\mathcal{L}(\mu)/\partial\mu = 0$ leads to

$$\hat{\mu} = (1/D)\sum_{d=1}^{D} \gamma_d. \tag{17}$$

Finally, the terms containing Σ are

$$\begin{aligned} \mathcal{L}(\Sigma) &= \sum_{d=1}^{D}\Big\{(1/2)\log\,|\sigma^{-1}| - (1/2)\mathrm{Tr}\left[\mathrm{diag}(\nu_d^2)\ \Sigma^{-1}\right] \\ &\quad -(1/2)(\gamma_d - \mu)^T\Sigma^{-1}(\gamma_d - \mu)\Big\}. \end{aligned} \tag{18}$$

Setting $\partial\mathcal{L}(\Sigma)/\partial\Sigma = 0$, leads to

$$\hat{\Sigma} = (1/D)\sum_{d=1}^{D}\left(I \cdot \nu_d^2 + (\gamma_d - \hat{\mu})(\gamma_d - \hat{\mu})^T\right). \tag{19}$$

Note that, once we obtained variational parameters $\{\gamma_d, \nu_d, \phi_d, \lambda_d, \zeta_d\}_{d=1}^{D}$ for all the training samples, we can update the model parameters $\{\mu, \Sigma, \pi, \beta\}$ by plugging them into Equation 10, 13, 17 and 19 until the overall likelihood in Equation 11 converges.

C. Inference on new image

Once we get variational and model parameters in hand, it remains to predict annotations from an unknown image. The first step is to perform variational inference for the unknown image, by fixing model parameters which have already obtained from training set.

The procedure for predicting annotations is the same as in [1]. We approximate the conditional distribution over codewords as follows:

$$p(w|\mathbf{v}) \approx \sum_{m=1}^{M}\sum_{z_m} q(z_m|\phi_m)p(w|z_m, \beta). \tag{20}$$

The conditional probability $p(w|\mathbf{v})$ can be treated as the predicted confidence score of each annotation word in word vocabulary, given the wholly codewords of the unknown image. Then we can choose the most promising annotations by ranking the confidence scores.

IV. Experimental Result

A. Configuration

We consider our models with two publicly available datasets that have been used in previous work: PASCAL VOC07 [13] and Corel5K [14]. Table I summarizes some statistics of these datasets.

- **PASCAL VOC07**. It contains around 10,000 images from Flickr, which are categorized in 20 different object classes and then annotated. Keeping the tags that at least appear 5 times, we obtained a cropped

TABLE I. STATISTICS OF TWO DATASETS: PASCAL VOC07 AND COREL 5K

	PASCAL VOC07	Corel 5K
Vocabulary size	186	260
Num. of training	4340	4500
Num. of testing	1860	499
Words per image (average / maximum)	5.4 / 28	4.7 /15
Image per word (average / maximum)	31.5 / 585	58.5 / 1004

dataset contains 6200 images with word vocabulary size 186 totally.

- **Corel 5K**. This dataset is an important and widely used benchmark for image retrieval and annotation. It contains around 5000 images manually annotated with 1 to 5 keywords. The word vocabulary contains 260 words. As Corel 5K is a copyright dataset, we just use the public statistical data[1] and evaluate the annotation performance.

While there exists a great variety of image features in the literature to choose from, and in this paper our purpose is to compare model rather than feature selection, thus we follow [4] and [7] to restrict our choice to SIFT feature, which has been shown to be discriminative in numerous classification and recognition problem. We use the public tool VLFeat[2] to extract 128-dimensional dense SIFT descriptor for each gray-scale image. We build a codeword dictionary for PASCAL VOC07 with size 500, for Corel 5K with size 1000. To train the model, we set the stopping criteria for log likelihood change is less than 0.001%, and max iteration 20 for variation inference, 100 for parameter estimation.

B. Test set likelihood

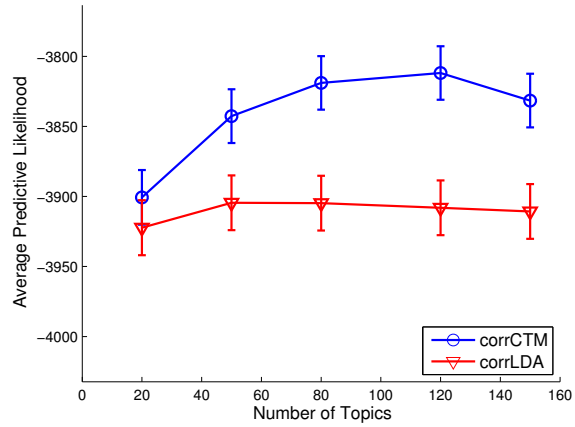
We trained a number of models for both corrCTM and corrLDA by changing values of topic number K on the two datasets, to compare the generalization performance of these two models. We compute the per-image average log likelihood of the hold-out test set. A model which better fits the data will assign a higher likelihood to the test set (for the log likelihood, lower value in abstract is better).

From Fig. 2, as expected, corrCTM provides a better fit than corrLDA for both two datasets on the held-out test data for all values of K. Compared with corrLDA, corrCTM gains 1.76% on PASCAL VOC07 and 4.45% on Corel 5K, for per-image average likelihood. Furthermore, corrCTM supports more topics, held-out likelihood of corrLDA peaks nearly at 50 topics on PASCAL VOC07, 80 topics on Corel 5K, while the likelihood for corrCTM peaks closely at 120 topics on both datasets. This indicates that corrCTM is more robust to over fitting when changing topic number than corrLDA.

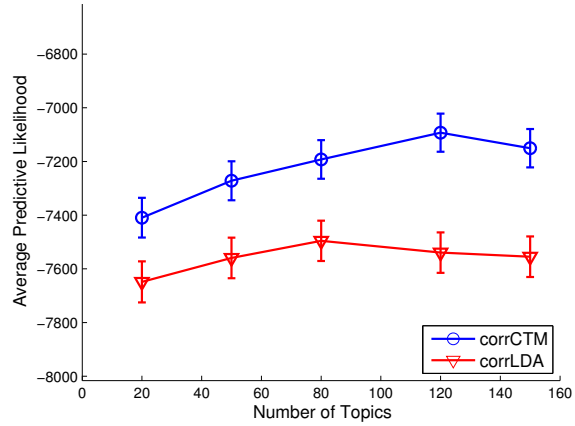
C. Predictive Perplexity

To measure the quality of codewords and annotations predicted by the models, we follow [10] and adopt "*feature perplexity*" and "*word perplexity*" as a performance measurement. Introduced from conventional language modeling, the perplexity is equivalent to the inverse of the geometric mean

[1]http://lear.inrialpes.fr/people/guillaumin/data.php
[2]http://www.vlfeat.org/



(a) PASCAL VOC07



(b) Corel 5k

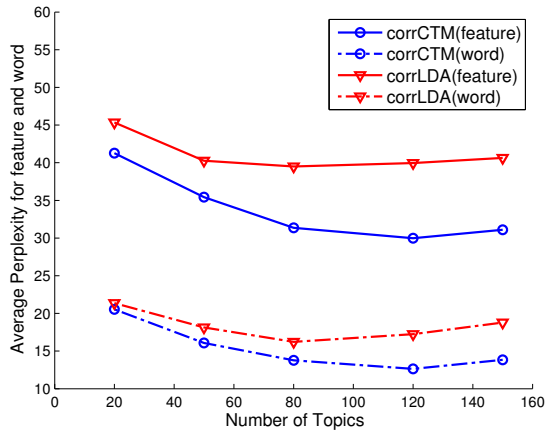
Fig. 2. The average held-out likelihood: (a)PASCAL VOC07 (b)Corel 5K

likelihood (in Equation 21 and 22, denoted as $\mathcal{P}(*)$), a lower perplexity score indicates better predictive performance),

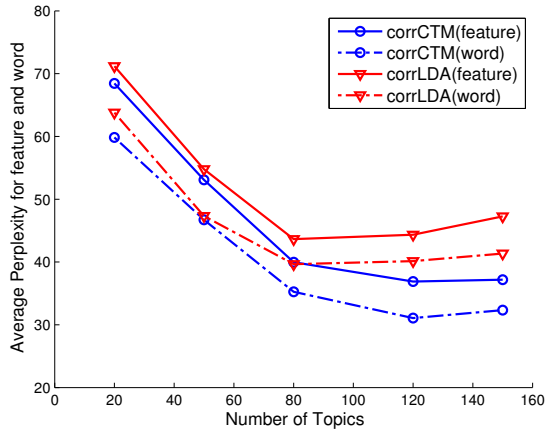
$$\mathcal{P}(feature) = \exp\left(-\frac{\sum_{d=1}^{D}\sum_{m=1}^{M_d}\log\, p(v_m|z_m)}{\sum_{d=1}^{D} M_d}\right), \quad (21)$$

$$\mathcal{P}(word) = \exp\left(-\frac{\sum_{d=1}^{D}\sum_{n=1}^{N_d}\log\, p(w_n|\mathbf{v}_d)}{\sum_{d=1}^{D} N_d}\right). \quad (22)$$

From Fig. 3(b), we see that corrCTM gains nearly 9% (resp. 7%) reduction on feature perplexity and 11% (resp. 6%) reduction on word perplexity on Corel 5K (resp. PASCAL VOC07). It is expected that the topic assignments for each feature are correlated according to the prior covariance structure among topics, then in the word indexing procedure, word will be selected according to the propagation of topic correlation, which leads a lower perplexity for word in corrCTM compared with corrLDA. In addition, like the overfit condition in held-out likelihood in Fig. 2, corrCTM also incorporates larger topic numbers (PASCAL VOC07: 50 *vs.* 120, Corel 5k: 80 *vs.* 120) than corrLDA for lower perplexity, which indicates corrCTM has better potentiality and more flexible to represent the occurrence of both features and words.



(a) PASCAL VOC2007



(b) Corel 5k

Fig. 3. Average predictive perplexity for feature and word on datasets: (a)PASCAL VOC07 (b)Corel 5K

D. Visualizing topic correlation

Since topic is able to decompose image-word relation to image-topic and word-topic relation in corrCTM. We are interested in examine how the proposed corrCTM better propagates the correlation to word-topic. Taking dataset Corel 5K for example, we first establish the normalized co-occurred frequency matrix for pair-wise words in vocabulary dictionary, and then obtain empirical frequently co-occurred word pairs by setting a threshold to the matrix. Finally, we get 1434 word pairs from the total 260 words, given a global threshold 0.25 which means when one word occurs, its partner would also occur with conditional probability larger than 0.25.

Our goal is to examine how well proposed corrCTM fits the empirical co-occurred word pairs. According to the caption-topic multinomial matrix β in Fig. 1(b), which is a $V_t \times K$ matrix and each column implies the multinomial distribution among words for current topic. We select top 10 caption words as most representative for each topic, and then make a statistic to recall the count of the appeared co-occurred word pair in the empirical co-occurred word pairs. We report the recall count for both datasets compared with corrLDA (#recall count / #total empirical count) in Table II. And Table III shows some typical co-occurred word pairs in top-10 words for topics in Corel 5K, here the total topic number K is 80, different word

TABLE II. RECALL COUNT FOR CO-OCCURRED WORDS: CORRCTM *vs.* CORRLDA

	PASCAL VOC 2007	Corel 5K
corrCTM	232 / 621	348 / 1434
corrLDA	185 / 621	276 / 1434

pairs from various semantic meaning are marked with different symbols (†, ‡ and §). It is indicated that corrCTM owns the ability to cluster more related word pairs to formulate compact topic representation than corrLDA, as it is intuitive that from the generative procedure of corrCTM, once correlated topic proportions are assigned to image features, the co-occurred words to related topics may also be probable. As a result, probability value for co-occurred word pairs would be lifted in the topic-word multinomial β.

E. Annotation Evaluation

For the annotation evaluation, we take advantages of standard performance measures used in image annotation community.

- **Precision and recall for fixed annotation length.** Following [7], each image is annotated with 5 most relevant keywords. Then we compute mean precision **P** and recall **R** over keywords of all images. In addition, **N+** denotes the number of keywords recalled non-zero times by model.
- **Mean average precision (mAP)** After computing the average precision for each keyword, **mAP** over all the keywords can be acquired.

Table IV compares our proposed corrCTM with corrLDA on the two datasets by setting their best topic number (for PASCAL VOC07, 120 *vs.* 50, for Corel 5K, 120 *vs.* 80). We find that our model improve over corrLDA for all the measures: 3% (resp. 4%) on **P**, 5% (resp. 3%) on **R**, 12 (resp. 16) on $N+$, and 4% (resp. 3%) on **mAP** on PASCAL VOC07 (resp. Corel 5K).

Examples of predicted annotations generated by corrCTM and corrLDA are shown comparatively in Fig. 4. We just use typical images in dataset PASCAL VOC07, as we have no right to use raw images from copyright Corel 5K dataset.

Top 4 images are satisfying examples predicted by our model, which correctly describe image content. In all these scenarios, annotations predicted by our model contain more correlated word pairs, such as (*bike*, *racing*), (*bird*, *natural*), (*desk*, *chair*), (*cat*, *kitty*), which are semantically related to the image content and ground truth captions, while corrLDA prefers more general words and less correlated word pairs. This superiority is reasonable because when modeling the topic correlation, word pairs that co-occur in one image will reinforce their multinomial probability values in the topic-word matrix β. In particular, our model has some potentiality to restrain mutually-exclusive word pairs (e.g. (*cat*, *pet*), (*puppy*, *pet*)), which is ignored in corrLDA. Bottom 2 images are improper annotation examples of proposed model. Although proposed model owns the ability to propagate correlation among words, it might suffer from "*error propagation*," in these 2 instances, once corrCTM has predicted some proper words *bus*, *sky*, it will still give high prediction score to

TABLE III. TYPICAL CO-OCCURRED WORD PAIRS IN TOP-10 REPRESENTATIVE WORDS OF EACH TOPIC: (**TOP PANEL**)CORRCTM AND (**BOTTOM PANEL**)CORRLDA

Topic index	Typical word pairs
Topic 1	cars†, prototype†, tracks†, street, turn, marsh‡, roofs, bengal§, forest‡, tiger§
Topic 4	plane†, jet†, sky†, sun, birds§, fly§, clouds§, snow, sand†, dunes†
Topic 27	snow†, ice†, polar†, frozen†, bear, mountain§, water, rocks§, grass, sky
Topic 48	island†, beach†, sand, sea†, water†, sky, people, kauai†, sunset, buildings
Topic 72	ocean†, coral†, fish†, rocks§, reefs§, water, orchid, boat§, sky, fan
Topic 1	water, sky†, tree, people, clouds†, grass, mountain, buildings, sun, snow
Topic 9	sky†, jet†, plane†, mountain, tree, water, sun, people, clouds, buildings
Topic 30	tree†, grass†, flowers§, people, field, house, mountain, sky, water, garden§
Topic 41	ice†, people, mountain§, sky, frost, snow§, clouds, water, rocks§, landscape
Topic 67	cars, buildings§, street†, people, sidewalk, lights†, window†, post, store‡, shops‡

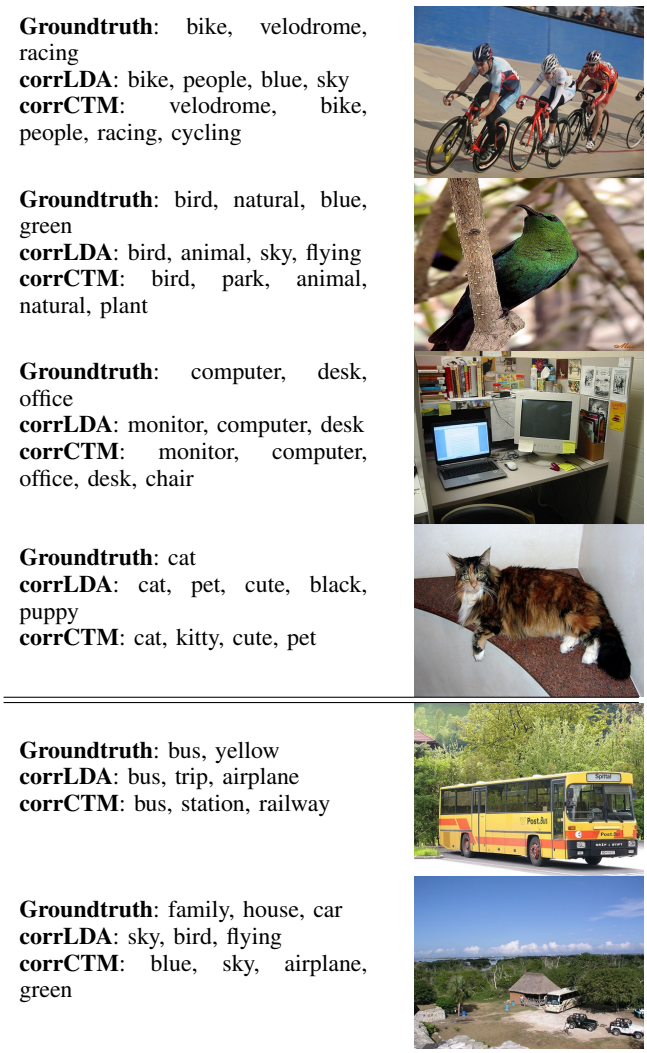


Fig. 4. Predicted annotations for example images from PASCAL VOC07. Top 4 examples are good predicted annotations by our model. Bottom 2 examples are incorrect predicted annotations by our model.

incorrect words such as *station*, *railway* and *airplane*, although they are inappropriate word-pairs to these images. Therefore, further effort to restrict topic correlation among words should be made to stop the error propagation in this condition.

TABLE IV. ANNOTATION PERFORMANCE: CORRCTM *vs.* CORRLDA

Dataset	Models	**P**	**R**	N+	**mAP**
PASCAL VOC07	corrLDA	17.8	27.9	49	19.1
	corrCTM	21.3	32.8	61	23.2
Corel 5K	corrLDA	24.9	35	53	24.6
	corrCTM	29.3	38.2	69	28.1

V. CONCLUSION

In this paper, we propose a novel statistical topic model corrCTM for the task of image annotation. We use logistic normal distribution instead of conventional Dirichlet assumption to capture the topic correlations. Topics are correlated with each other to mix the topic proportions for one image. The presence/absence of image features and annotation words are later influenced and generated given correlated topic proportions. Experiments on 2 benchmark datasets indicate that modeling topic correlation not only leads to a better topic representation fit for image features and annotation words, but also perform superior annotation quality compared with LDA based corrLDA method. In addition, we conclude that other LDA based methods, mentioned as [4] [6], would also benefit from modeling topic correlation, because they share the same methodology of topic representation in generative process. For the future work, we will develop more flexible model to propagate topic correlations from image features to annotation words, to generate more compact correspondence between features and words through topic space.

REFERENCES

[1] D.M.Blei and M.I.Jordan, "Modeling annotated data," in *In Proceedings of the 26th annual international ACM SIGIR conference on Research and development in information retrieval*, 2003, pp. 127–134.

[2] D.M.Blei and J.D.Lafferty, "Correlated topic models," in *Neural Information Processing Systems (NIPS)*, 2006.

[3] D. Grangier and S. Bengio, "A discriminative kernel-based model to rank images from text queries," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2008.

[4] C.Wang, D.M.Blei, and F. fei Li, "Simultaneous image classification and annotation," in *International Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

[5] G.Carneiro and N.Vasconcelor, "A model for learning the semantics of pictures," in *Neural Information Processing System (NIPS)*, 2003.

[6] D. Putthividhya, H. T.Attias, and S. S.Nagarajan, "Topic regression multi-modal latent dirichlet allocation for image annotation," in *International Conference on Computer Vision and Pattern Recognition (CVPR)*, 2010.

[7] D.Putthividhya, H.T.Attias, and S.S.Nagarajan, "Supervised topic model for automatic image annotation," in *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2010.

[8] D.M.Blei and J.D.McAullffe, "Supervised topic models," in *Neural Information Processing System (NIPS)*, 2007.

[9] T. J. Malisiewicz and J. C. Huang, "Detecting objects via multiple segmentations and latent topic models," Carnegie Mellon University, Tech. Rep., 2006.

[10] D.M.Blei, A.Y.Ng, and M.I.Jordan, "Latent dirichlet allocation," *Journal of Machine Learning Research*, pp. 993–1022, 2003.

[11] D.M.Blei and J.D.Lafferty, "A correlated topic model of science," *Annals of Applied Statistics*, 2007.

[12] S. J. Wright, *Numerical Optimization*, 2nd ed. Berlin, Heidelberg: Springer Verlag, 2006.

[13] M. Everingham, L. VanGool, C. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes challenge 2007 (voc2007) results," http://www.pascal-network.org/challenges/VOC/voc2007/workshop/index.html.

[14] P. Duygulu, K. Barnard, J. de Freitas, and D. Forsyth, "Object recognition as machine translation: Learning a lexicon for a fixed image vocabulary," in *ECCV*, 2002, pp. 97–112.